Running SIRISYS on your Mac

A step-by-step guide from scratch for STRATA, an independent software interpretation of SIRISYS (version 13.4). No programming needed, and no prior use of the Terminal.

Before you start

You will need:

- Your Mac and an internet connection.
- The **STRATA** repository on GitHub: **github.com/juanan-prz/STRATA** (step 1 explains how to download it).
- For the experiments with a model: **Ollama**, free and local (recommended), or an Anthropic API key (optional). The functional tests need neither.

The first time takes about 30 minutes of installation. After that, each session starts in a minute.

> **Golden rule:** copy and paste the commands exactly as they are, without changing anything. One extra space or a misplaced character and they fail. Copying and pasting is the right way, not cheating.

Step 1 · Download the project

- Open **https://github.com/juanan-prz/STRATA** in Safari.
- Click the green **Code** button and then **Download ZIP**.
- Open the downloaded ZIP (double-click): a folder appears.
- Move it to **Documents** and rename it to exactly **sirisys** (lower case, no accents or spaces).

Inside, together and with no subfolders, you should find these files (among others):

File	What it is for
`Sirisys_Framework_v13_4.py`	The system.
`Sirisys_Server.py` · `sirisys_loader.py`	Start experiment mode, with the live view.
`Sirisys_Live_Visualizer.html`	The live view of the field.
`Sirisys_Static_Visualizer.html`	The inspector: examines a saved instance afterwards.
`run_report.py`	Measured runs: writes a report with their results.
`field_topology.py` · `sirisys_measures.py`	Measurements the reports use. They also check that the model answers.
`test_*.py` · `sirisys_testkit.py`	Functional tests: five files and what they share.
`measure_phase3.py` · `live_c2.py` · `soak.py`	Measurements: criteria, real percepts and a long life.
`requirements.txt`	The list of pieces to install.

Always download on the computer, not on a phone. If the files pass through a phone's clipboard or a rich-text editor, the quotation marks get corrupted and the code stops working.

When a new version of the framework comes out, just put it in this folder: the programs always use the highest version, and experiment mode and the reports say which one they loaded.

Step 2 · Open the Terminal

The Terminal is a window where you type commands instead of clicking. Press **Command + Space**, type **Terminal** and press Enter. A window opens with a blinking cursor. You cannot break anything by typing there: if a command does not exist, it simply says so.

Step 3 · Check whether you have Python

Copy this into the Terminal and press Enter:

```
python3 --version
```

If it answers **Python 3.10** or higher (for example 3.12.4), you have it: skip to step 4. If it says command not found, offers to install developer tools, or the version is older: go to **python.org/downloads**, click the yellow **Download Python** button, install it like any other application, close the Terminal, open it again and repeat the command.

Step 4 · Go to the folder

The Terminal needs to know where the files are. Type **cd** followed by a space (do not press Enter), drag the **sirisys** folder from the Finder onto the Terminal window and, when the path appears, press Enter. To check:

```
ls
```

It should list the files from step 1. If you see them, you are on track.

Step 5 · Install the pieces

```
pip3 install -r requirements.txt
```

You will see a lot of text scroll by: that is normal and takes a couple of minutes. When you can type again, it has finished. If a red warning says externally-managed-environment (it usually happens with Homebrew's Python), use this line instead:

```
pip3 install --break-system-packages -r requirements.txt
```

Step 6 · Check that everything works (free, no model)

Run these five tests, one at a time. They cost nothing and need no internet:

Command	What it should say at the end
`python3 test_sirisys.py`	SUMMARY: 24/24 passed (≈ 15 s)
`python3 test_phase1.py`	SUMMARY: 51/51 passed (≈ 15 s)
`python3 test_phase2_guarantee.py`	SUMMARY: 6/6 passed (≈ 5 s)
`python3 test_phase4.py`	SUMMARY: 12/12 passed (≈ 5 s)
`python3 test_ontology.py`	SUMMARY: 16/16 passed (≈ 5 s)

If any line says **FAIL**, note which test failed and the lines before it; download the ZIP again on the computer and repeat, since a damaged file is the most common cause. To see the field live:

```
python3 Sirisys_Server.py --play
```

Open **http://127.0.0.1:8000** in Safari. To stop it, go back to the Terminal and press **Control + C**: the instance is saved in the **sirisys_runs** folder and nothing is deleted. To examine it afterwards, open **Sirisys_Static_Visualizer.html** in Safari and choose the **state.json** in its folder.

Step 7 · Choose the model

Option A — Ollama (recommended). The model runs inside your Mac: free, and nothing is sent out.

- Download it from **ollama.com**, install it like any other application and open it (an icon stays in the menu bar).
- In the Terminal, download a model. For example:

```
ollama pull llama3.1
ollama list
```

ollama list should show the downloaded model. Always use the same model in the runs you intend to compare.

Option B — Anthropic (optional, for experiments only). At **console.anthropic.com** create an account, go to **API Keys → Create Key**, copy the key (it starts with sk-ant-; it is shown only once) and add credit under **Billing** without turning on automatic reload. A Claude subscription does not include API credit: they are separate. Then, in the Terminal:

```
export ANTHROPIC_API_KEY="sk-ant-your-key-here"
```

It answers nothing: that means it worked. It lasts while the window stays open. Never store the key in a file in the folder.

Step 8 · The short report test

```
python3 run_report.py --cycles 20 --seeds 1 --backend ollama --profile minimal
```

(With Anthropic, change **ollama** to **anthropic**.) Before it starts, the program asks the model two real questions. Look at:

What to look for	What it should say
The 14 check lines at the start	All with [ok].
model probe	ok, and the seconds each call takes. If the model does not answer or answers empty, the program stops here with **STOP** and says why (see «If something fails»).
sample names	Two names made up by the model, in the framework's language.
At the end: model calls	failures 0 and empty 0.

Step 9 · Suggested experiments

A sequence that compares the field with and without a model. All with the **minimal** profile (no narration, no time of day), in this order. The long ones start with **caffeinate -i**: it keeps the Mac awake while the test runs and stops on its own when it ends. Do not close the Terminal window; on a laptop, keep it plugged in with the lid open.

1 · Real percepts, no model (≈ 10 minutes). The reference: the field with your Mac's percepts over a long life. **Do not use the Mac while it runs**: the field perceives its processor and its memory.

```
python3 live_c2.py 10000 none minimal > live_none.json
```

2 · The model against a control without a model (several hours). Three arms on the same three percept streams: the model, the model asking only the Sombunal questions, and the control. At the start it says which model answers and the most hours it will take. You may use the Mac during this one.

```
caffeinate -i python3 run_report.py --backend ollama --profile minimal --cycles 1000 --seeds 3
```

3 · Real percepts with the model (several hours; a good one to leave overnight). At the start it says in the Terminal the most hours it will take. **Do not use the Mac while it runs.**

```
caffeinate -i python3 live_c2.py 3000 ollama minimal > live_ollama.json
```

4 · Optional, recommended: does the model always answer the same? (it takes a little over half as long as 2). It repeats the same run with the model twice and checks whether the field comes out identical. At the end it says **determinism: YES** or **NO**: both answers are useful, neither is an error. It writes its own file, so it does not delete the one from 2. You may use the Mac during this one.

```
caffeinate -i python3 run_report.py --backend ollama --profile minimal --cycles 1000 --determinism
```

While **live_c2.py** runs nothing else appears on screen: that is normal, it writes the result at the end. **run_report.py** prints a progress line every 25 cycles.

If the model stops answering halfway (Ollama closes or hangs), the program stops on its own and says so in the Terminal with **STOPPED**. Open Ollama and run that same command again; the interrupted file is not needed.

Optional: watch the field live with the model.

```
export SIRISYS_LLM_BACKEND=ollama
python3 Sirisys_Server.py --play --profile minimal
```

Open **http://127.0.0.1:8000** as in step 6. The model is shown at the top; if it stops answering, the field pauses, the header marks it in red (**watch tripped**) and the event log says why.

Step 10 · Your results

The **sirisys** folder will now hold **live_none.json**, **sirisys_report.json** and **live_ollama.json**, plus **sirisys_determinism.json** if you ran 4. They are text files (JSON): each records the run's configuration, the self-adjustment criteria and their verdict, the model's health and the timings; those of run_report.py also record each run's ledger head. The technical guide (**SIRISYS_Guide_v13_EN.md**) explains what each one contains. To compare results across machines or models, also note:

- the name of the Ollama model you used (**ollama list** shows it);
- your Mac's chip and memory (Apple menu → **About This Mac**).

If something fails

What you see	What to do
command not found: python3	Python is not installed: go back to step 3.
No such file or directory	You are not in the folder or files are missing: repeat step 4 and check with ls.
externally-managed-environment	Use the second line of step 5.
A test says FAIL	Download the ZIP again on the computer and repeat. If it still fails, note the test and its message.
model active: False · STOP: ... not available	Ollama is not open (open it and repeat), the model is not downloaded (ollama list) or, with Anthropic, the key was not set in this window.
STOP: ... answered empty	The model «thinks» before answering and runs out of room for the name. Use a general-purpose model, for example llama3.1 (step 7).
STOPPED — model health	The model stopped answering halfway. Open Ollama and run that command again.
authentication_error / 401 · credit balance is too low	With Anthropic: the key was copied wrong (copy all of it, without spaces) or there is no credit at console.anthropic.com → Billing.
already open by another writer	Another SIRISYS is writing in that folder: close the other Terminal window or press Control + C in it.
Address already in use	Experiment mode is already running in another Terminal window: stop it there with Control + C.

What you see	What to do
The test stopped when the Mac slept	Run it again with caffeinate -i in front; on a laptop, plugged in with the lid open.
The window seems stuck	Normal: it is working. If it really hung, Control + C.

Summary

Each session starts in the folder, and it is worth checking that everything works:

```
cd   (and drag the sirisys folder here)
python3 test_sirisys.py
python3 test_phase1.py
python3 test_phase2_guarantee.py
python3 test_phase4.py
python3 test_ontology.py
```

The sequence of experiments from step 9:

```
python3 live_c2.py 10000 none minimal > live_none.json
caffeinate -i python3 run_report.py --backend ollama --profile minimal --cycles 1000 --seeds 3
caffeinate -i python3 live_c2.py 3000 ollama minimal > live_ollama.json
caffeinate -i python3 run_report.py --backend ollama --profile minimal --cycles 1000 --determinism
```